Travel & Tourism Web Application

A

Minor Project

Submitted in partial fulfilment for the award of the degree of

BACHELORS OF TECHNOLOGY

IN

Computer Science Engineering

by

Name of the Students

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(April 2025)

CERTIFICATE

This is to certify that the project entitled "Smart Travel & Tourism Web Application" submitted by G. Kalyani (BETN1AI23057) and S. Ranga Swamy (BETN1AI23056) in partial fulfilment for the award of the degree of B.Tech. (Computer Science Engineering), ITM University, Gwalior, has been carried out under my/our supervision.

To the best of my knowledge and belief, the project report:

(i) Is an original piece of work of the candidates.

(ii) Has duly been completed.

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(v) Work has not been submitted partially or wholly to any other University or Institute for the award of this or any other degree or diploma.

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We hereby declare that the work entitled "Smart Travel & Tourism Web Application" submitted to the Department of Computer Science Engineering, School of Engineering and Technology, ITM University, Gwalior (M.P.) is our original work done under the supervision of Ms. Pragya Jain, Asst. Professor, Dept. of CSE. This project report does not contain any part which has been submitted for the award of any degree either in this University or in any other University.

We further declare that the work is free from any plagiarism.

Signature:	**Signature:**
G. Kalyani (BETN1AI23057)	S. Ranga Swamy (BETN1AI23056)

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Acknowledgement

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Abstract

Travel and Tourism is one of the fastest-growing industries in the world, contributing significantly to the global economy. The traditional method of booking tours involves visiting travel agencies manually, making phone calls, and managing paper-based records, which is time-consuming, error-prone, and lacks transparency.

This project presents the design and development of "Smart Travel & Tourism Web Application" — a full-stack web-based system developed using PHP, MySQL, HTML, CSS, and Bootstrap. The system allows users to browse tour destinations, register and login securely, and book holiday packages online.

The application includes modules for user registration and authentication, destination browsing, booking management, and an admin dashboard. The system uses a MySQL database (travel_system) for persistent data storage and ensures secure communication between the client and server.

The project demonstrates the practical implementation of a full-stack web application covering frontend design, backend logic, and database connectivity. Future enhancements include payment gateway integration, package reviews, and mobile application support.

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LIST OF ABBREVIATIONS

PHP — Hypertext Preprocessor
WAMP/XAMPP — Windows Apache MySQL PHP Stack
UI — User Interface
UX — User Experience
HOD — Head of Department
CSE — Computer Science Engineering
IT — Information Technology
CRUD — Create, Read, Update, Delete
URL — Uniform Resource Locator
API — Application Programming Interface
HTML — Hyper Text Markup Language
CSS — Cascading Style Sheets
MySQL — My Structured Query Language
DB — Database

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CHAPTER 1
INTRODUCTION & OBJECTIVES

1.1 Background

Travel and Tourism is one of the most dynamic and fast-growing sectors of the global economy. According to the World Travel & Tourism Council (WTTC), the tourism industry contributed approximately 10.4% to the global GDP before the COVID-19 pandemic. With the rapid advancement of information technology and the widespread use of the internet, online travel booking systems have transformed the way people plan and book their holidays.

In India, the Ministry of Tourism (Annual Report 2023-24) reports a consistent rise in both domestic and international tourist arrivals, indicating a strong demand for efficient and user-friendly online travel platforms. However, a large number of small and medium-sized travel agencies still operate manually or through basic static websites, lacking the features needed to serve modern travelers.

This project addresses the need for a comprehensive web-based Travel & Tourism System that enables users to browse destinations, register, login, and book tours online — all from a single digital platform.

1.2 Problem Statement

The traditional travel booking process involves visiting travel agencies in person, making phone calls to inquire about tour packages, and managing paper-based booking records. This approach suffers from several critical problems:

- Time-consuming and inconvenient for users who must visit agencies physically.
- Paper-based records are highly prone to errors, mismanagement, and data loss.
- No centralized platform for users to compare and browse multiple tour packages.
- Travel agencies struggle to manage bookings efficiently without a digital system.
- Lack of real-time information on tour availability and package details.

1.3 Objectives

The primary objectives of this project are as follows:

1. To design and develop a full-stack Travel & Tourism web application using PHP and MySQL.
2. To implement user registration and login with secure database-backed authentication.
3. To create a Destinations module where users can browse various travel packages.
4. To build a Booking module that allows users to book tour packages online.
5. To develop an Admin Dashboard for managing user bookings and destinations.
6. To design a responsive and professional frontend using Bootstrap 5.
7. To ensure proper database connectivity using PHP MySQLi extension.

CHAPTER 2
LITERATURE SURVEY

2.1 Existing Systems

Several online travel booking systems and platforms currently exist in the market. A review of the existing systems is presented below:

a) MakeMyTrip

MakeMyTrip is one of the leading online travel portals in India, offering flight bookings, hotel reservations, and holiday packages. While it is highly feature-rich, it requires significant infrastructure, third-party payment gateways, and real-time API integrations, making it unsuitable as a reference model for small-scale institutional projects.

b) Yatra.com

Yatra.com is another major travel booking platform in India. It supports user authentication, destination search, and tour package listing. However, similar to MakeMyTrip, it relies heavily on proprietary technologies and external APIs.

c) Static Travel Websites

Many small travel agencies maintain static HTML/CSS websites that display only basic information about their services. These websites do not support dynamic booking, user accounts, or database-driven content management.

2.2 Comparative Analysis

Feature	MakeMyTrip	Yatra.com	Static Sites	Proposed System
User Login/Register	✓	✓	✗	✓
Destination Browsing	✓	✓	Limited	✓
Online Booking	✓	✓	✗	✓
Admin Dashboard	✓	✓	✗	✓
Open Source / Academic	✗	✗	✓	✓

Database-driven	✓	✓	✗	✓

Table 4: Comparison of Existing Systems

CHAPTER 3
RESEARCH GAP & PROPOSED SYSTEM

3.1 Research Gap

Based on the literature survey and analysis of existing systems, the following research gaps have been identified:

- Most commercial travel platforms are too complex and costly to be deployed for small travel agencies or educational purposes.
- Lightweight, open-source, and database-driven travel booking systems suitable for academic mini-projects are largely absent.
- Existing static travel websites do not support any form of user interaction, booking, or account management.
- There is a lack of beginner-friendly full-stack travel systems built using PHP and MySQL that can serve as reference implementations.

3.2 Proposed System

The proposed system, "Smart Travel & Tourism Web Application," addresses all the identified research gaps by providing a complete, lightweight, and database-driven travel booking solution. The system is built using PHP, MySQL, HTML, CSS, and Bootstrap — technologies that are widely taught in B.Tech. CSE curriculum and require no external API subscriptions or third-party dependencies.

The proposed system includes the following components:

- Home Page (index.php): A responsive landing page with navigation, hero section, and quick links.
- User Authentication (login.php, register.php): Secure user registration and login using MySQL database.
- Destinations Module (destinations.php): A dynamic page displaying available tour destinations.
- Booking Module (booking.php): A booking form allowing users to select and confirm tour packages.
- Admin Dashboard (admin/dashboard.php): A protected admin panel for managing bookings.
- Database Layer (config/db.php): Centralized MySQL database connection management.

CHAPTER 4
IMPLEMENTATION

4.1 Tools & Technologies Used

S.No.	Technology	Purpose
1	PHP 7+	Backend server-side scripting language
2	MySQL	Relational database management for storing users and bookings
3	HTML5	Structuring web pages and forms
4	CSS3	Styling web pages (custom styles)
5	Bootstrap 5.3	Responsive frontend UI framework
6	XAMPP / WAMP	Local server environment (Apache + MySQL + PHP)
7	VS Code	Code editor / Integrated Development Environment
8	phpMyAdmin	Database management and query interface

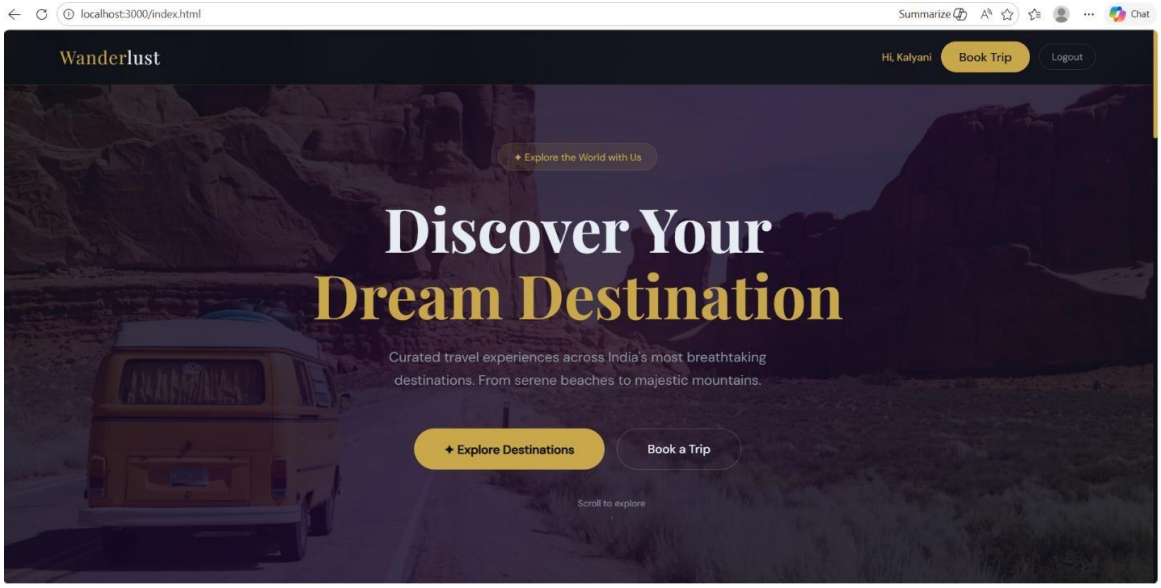
Table 1: Tools & Technologies Used

4.2 System Architecture

The system follows a three-tier architecture consisting of the Presentation Layer (HTML/CSS/Bootstrap frontend), the Application Layer (PHP backend scripts), and the Data Layer (MySQL database). The browser sends HTTP requests to the PHP server, which processes the request, queries the MySQL database, and returns a dynamic HTML response to the user.

System Architecture:

Browser (User) → Apache Web Server → PHP Scripts → MySQL Database



4.3 Modules Description

a) Home Page (index.php)

The home page serves as the landing page of the application. It contains a Bootstrap-based navigation bar with links to all major sections: Home, Destinations, Booking, Login, and Register. The hero section displays the tagline 'Explore The World — Discover Your Dream Destination' with call-to-action buttons.

b) User Registration (register.php)

The registration module collects the user's name, email, and password through an HTML form. The submitted data is stored in the users table of the MySQL database (travel_system). Basic server-side validation ensures no empty fields are submitted.

c) User Login (login.php)

The login module authenticates users by querying the MySQL database for matching username and password records. Upon successful authentication, the user is redirected to the admin dashboard. An error message is displayed for invalid credentials. Note: future enhancement will include password hashing using PHP's password_hash() function.

d) Destinations (destinations.php)

The destinations module displays various tour packages and travel destinations. It is designed to fetch destination data from the database and present it in a card-based Bootstrap layout.

e) Booking (booking.php)

The booking module allows users to select a destination and submit a booking request. It collects user details, preferred travel dates, and number of travelers. The booking data is stored in the bookings table of the database.

f) Admin Dashboard (admin/dashboard.php)

The admin dashboard is a protected page accessible only after authentication. It provides an overview of all bookings and user registrations, enabling the administrator to manage the system effectively.

4.4 Database Design

The database name is travel_system and is hosted on MySQL localhost. It consists of the following main tables:

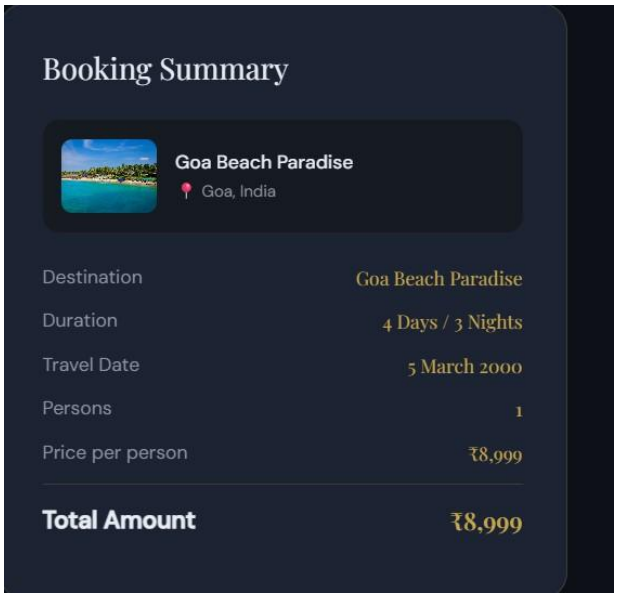
Table 2: users

Field Name	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique user ID
username	VARCHAR(100)	NOT NULL	User login name
password	VARCHAR(255)	NOT NULL	User password
email	VARCHAR(150)	NOT NULL	User email address

Table 3: bookings

Field Name	Data Type	Constraints	Description
booking_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique booking ID
user_id	INT	FOREIGN KEY (users.id)	Reference to user
destination	VARCHAR(200)	NOT NULL	Selected destination
travel_date	DATE	NOT NULL	Preferred travel date
travelers	INT	NOT NULL	Number of travelers

status	VARCHAR(50)	DEFAULT 'Pending'	Booking status



CHAPTER 5
RESULTS & DISCUSSIONS

The Smart Travel & Tourism Web Application was successfully developed and tested on a local XAMPP/WAMP server environment. All the planned modules were implemented and verified for their intended functionality.

5.1 Testing Results

Module: Home Page (index.php)

Result: Rendered correctly with Bootstrap navbar and hero section. Navigation links work as expected.

Module: User Registration (register.php)

Result: Successfully stores new user credentials in the MySQL users table.

Module: User Login (login.php)

Result: Correctly authenticates users and redirects to admin dashboard on success.

Module: Destinations Page (destinations.php)

Result: Displays tour destinations correctly in responsive card layout.

Module: Booking Module (booking.php)

Result: Booking form submission successfully stores records in the bookings table.

Module: Admin Dashboard (admin/dashboard.php)

Result: Protected admin panel loads successfully post-authentication.

Module: Database Connectivity (config/db.php)

Result: Stable connection established to MySQL database 'travel_system'.

5.2 Discussions

The system successfully meets all its defined objectives. The PHP backend correctly processes form submissions, communicates with the MySQL database, and returns appropriate responses to the browser. The Bootstrap-based frontend ensures responsive behavior across devices.

One identified limitation in the current implementation is the absence of password hashing in the login/register modules, which uses plain-text password comparison. This will be addressed in future versions using PHP's

password_hash() and password_verify() functions. Additionally, the booking module requires integration with a payment gateway for real-world deployment.

CHAPTER 6
CONCLUSIONS & FUTURE WORK

6.1 Conclusion

The Smart Travel & Tourism Web Application was successfully designed, developed, and tested as a part of the B.Tech. Minor Project requirement. The project demonstrates a practical implementation of a full-stack web application covering all key aspects of modern web development — frontend design, backend scripting, and database management.

The system provides users with a clean interface to browse tour destinations and book packages online, solving the key problems associated with manual, paper-based travel booking processes. The admin dashboard enables effective management of all user registrations and bookings.

All six planned objectives of the project were successfully achieved during implementation and testing.

6.2 Future Work

The following enhancements are planned for future versions of the system:

- Integration of a secure payment gateway (e.g., Razorpay, PayPal) for real online transactions.
- Implementation of password hashing using PHP's password_hash() and password_verify() functions.
- Addition of a user profile page allowing travelers to view and manage their past bookings.
- Integration of email notifications for booking confirmations using PHP Mailer.
- Development of a mobile application (Android/iOS) using the existing PHP backend as an API.
- Introduction of package reviews and rating system for destinations.
- Deployment to a live web server (e.g., Hostinger, AWS) with a registered domain.

REFERENCES

[1] PHP Official Documentation — https://www.php.net/docs.php

[2] MySQL Reference Manual — https://dev.mysql.com/doc/

[3] Bootstrap 5.3 Documentation — https://getbootstrap.com/docs/5.3/

[4] W3Schools PHP Tutorial — https://www.w3schools.com/php/

[5] Ministry of Tourism, Government of India — Annual Report 2023-24

[6] World Travel & Tourism Council (WTTC) — India 2024 Report

[7] MDN Web Docs — HTML, CSS, JavaScript Reference — https://developer.mozilla.org

[8] R. Nixon, "Learning PHP, MySQL & JavaScript", O'Reilly Media, 6th Edition, 2021.

[9] phpMyAdmin Documentation — https://docs.phpmyadmin.net/

[10] Unsplash.com — Free travel images reference — https://unsplash.com